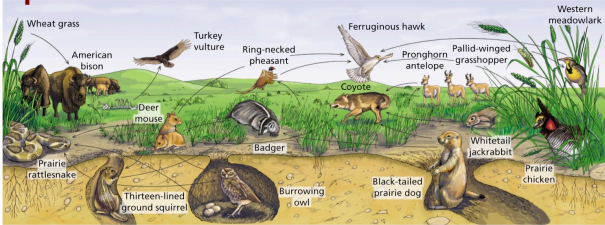


Chapter 52 Objectives: Be able to...

- Describe and recognize the **four primary types** of community interactions.
- Contrast **realized and fundamental niches**.
- Explain why competition can lead to **character displacement** and **resource partitioning**.
- Differentiate between 3 different types of **consumption**.
- Explain the **defensive strategies of prey species**.
- Explain how interactions lead to **coevolution**.
- Describe the roles and some examples of **keystone species**.
- Define **succession** and differentiate between **primary and secondary**.
- Describe the pattern of **species invasion** during succession.

Community ecology: The study of species interactions



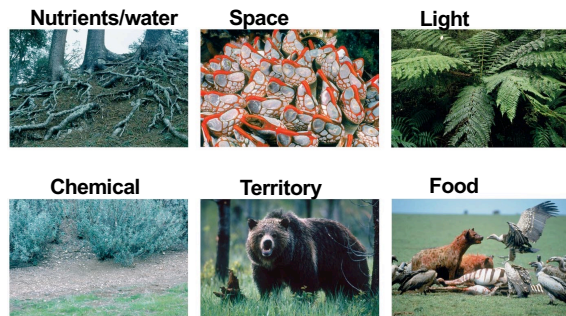
Community:

Through interactions, organisms co-evolve within communities

Interaction Species 1 Species 2

- Competition
- Consumption
- Mutualism
- Commensalism

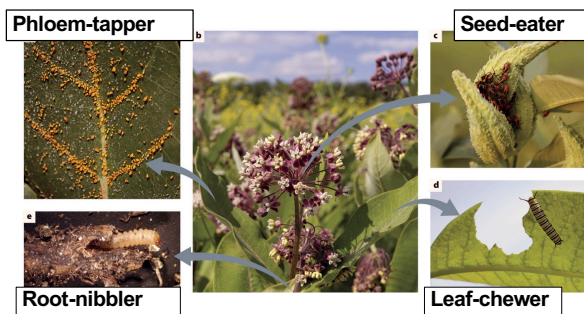
Competition (-/-):

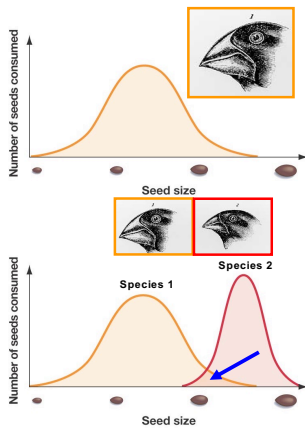


Ecological niche:

Niche	Placental Mammals	Australian Marsupials
Burrower	Mole	Marsupial mole
Anteater	Lesser anteater	Numbat (anteater)
Mouse	Mouse	Marsupial mouse
Climber	Lemur	Spotted cuscus
Glider	Flying squirrel	Flying phalanger
Cat	Ocelot	Tasmanian "tiger cat"
Wolf	Wolf	Tasmanian wolf

Niche-





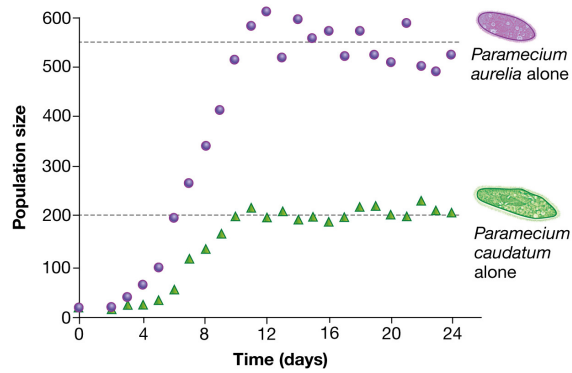


Fig. 51.8

Consequences of competition:

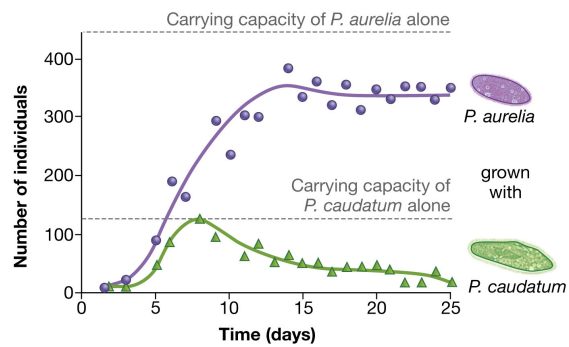
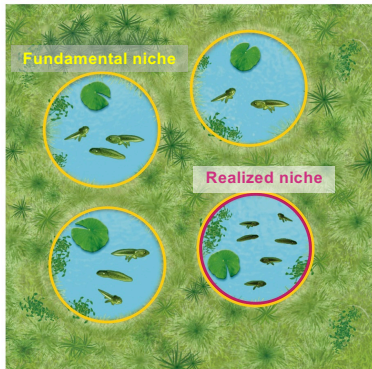


Fig. 52.2

Consequences of competition:



Fundamental Niches

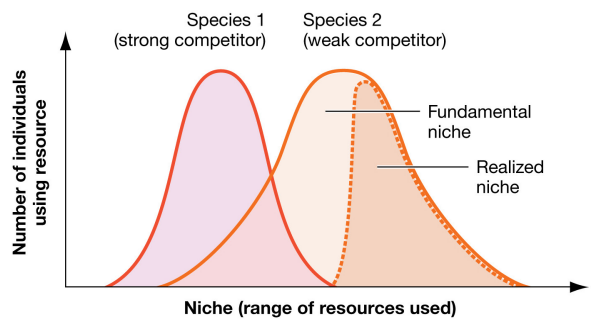
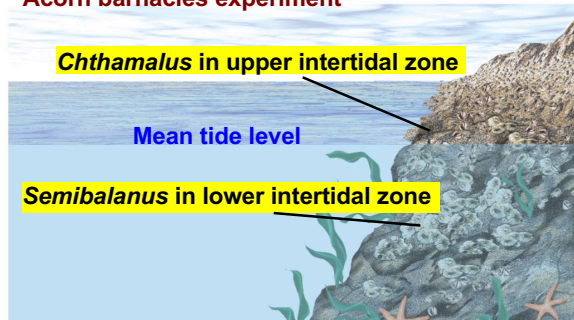
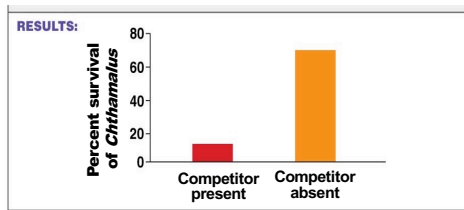
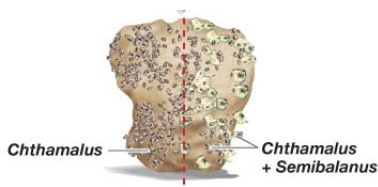


Fig. 52.3

Acorn barnacles experiment



Why no *Chthamalus* in lower?



Competition Can Lead to Niche Differentiation

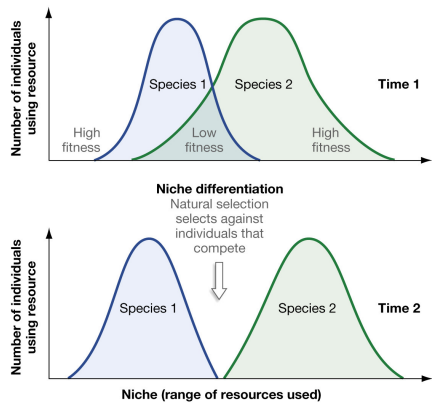
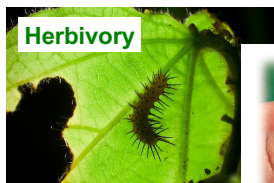


Fig. 52.4

Consumption (+/-): one organism eats another



Constitutive Defenses Are Always Present



Cryptic coloration



Escape Behavior



Toxins and other
defense chemicals



Schooling/flocking



Defense armor
and weapons

Mimicry

Müllerian mimics:



Paper wasp



Bumblebee



Honeybee

Batesian mimics:



Hornet moth



Wasp beetle



Hoverfly

Cool parasite coevolution story...

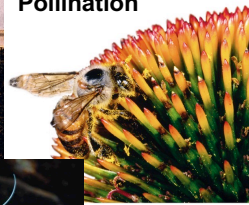


Mutualism (+/+):

Mycorrhizae



Pollination



Coevolution between *Acacia* & ants

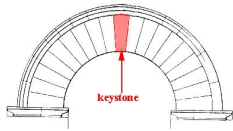


Commensalism (+/0):



Fig 45.11

Keystone species



Predator: *Pisaster ochraceus*



Prey: *Mytilus californianus*

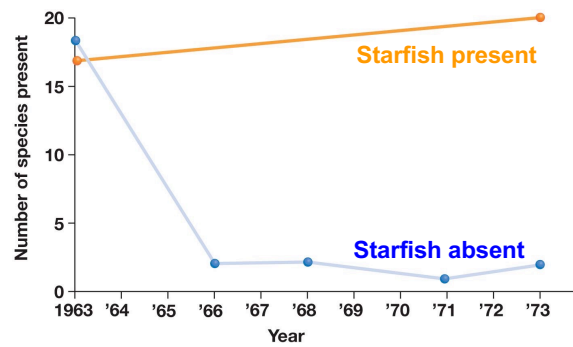
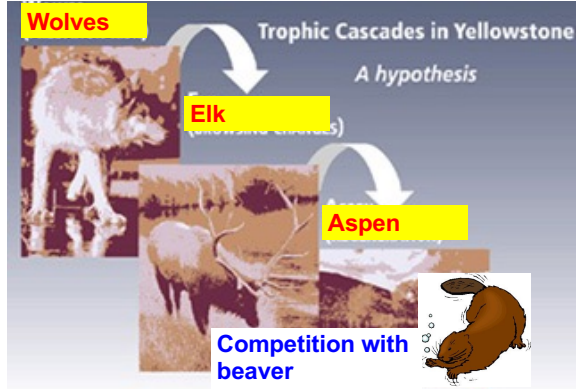


Fig. 52.11

Wolf: Another keystone predator



Wolves indirectly help beaver by decreasing elk



Wolves indirectly help beaver by decreasing elk



Yellowstone Wolves





<https://www.fs.usda.gov/>



Succession:

Primary succession:



Disturbance:





Old field



Disturbance ends



Pioneer species



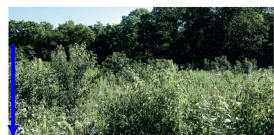
Pioneer species replaced by longer-lived competitors



Early successional community



Shrubs and shade intolerant trees invade.



Mid-successional community

↓
Long-lived, competitive, shade-tolerant
trees invade in understory.



**Late-successional
community**



Climax community
